MultiSIM 10 and 11 Circuits to Accompany

Electronic Devices, 9th Edition by Tom Floyd

Notes and Clarifications

Effective 20 Aug 10

1. Overview

MultiSIM 10 and 11 Circuits to Accompany Electronic Devices, 9th Edition by Tom Floyd consists of 2 separate sets of files. Student circuit files can be found on the CD packaged with the course textbook. Circuit files and solution files for each circuit file are available for the instructor. These can be found on the ITT Instructor Resource Center website. See your courseware for location of and access to these files.

2. CD Organization

Files on the Student CD are organized as shown in Figure 1. Although the figure below illustrates the organization for the MultiSIM 11 folders, the organization of the MultiSIM 10 folders is similar.

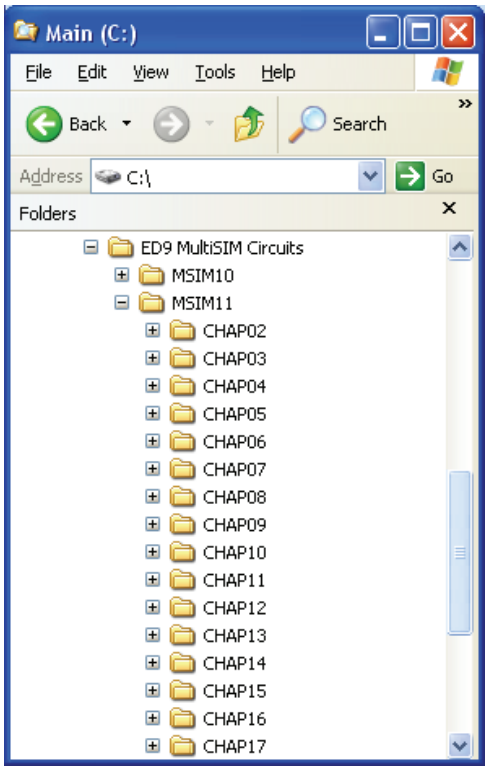


Figure 1: Sudent CD Folder Organization

Files for the instructor are organized as shown in the following folders. The MSIM10 folder contains the same subfolders as those shown for the MSIM11 folder.

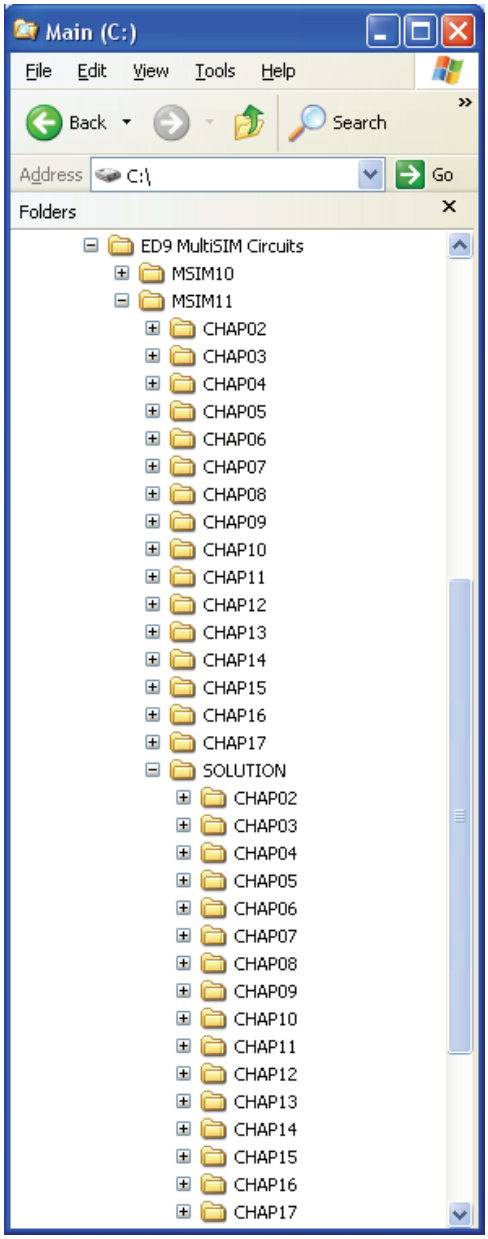


Figure 2: Instructor Folder Organization

2.1. Circuit File Organization

Each CD contains files in both Multisim 10 and Multisim 11 formats. All Multisim 10 circuit files are contained in the chapter subfolders in MSIM10 folder and have a .MS10 file extension. All Multisim 11 circuits are contained in the chapter subfolders in the MSIM11 folder and have a .MS11 file extension. Multisim 10 will work only with the .MS10 and earlier file formats, and Multisim 11 will work with .MS11 and earlier file format, include .MS10 files. Note, however, that Multisim 10 and Multisim 11 can save files only in their "native" file formats.

2.2. Solution File Organization

Solution files are contained in the chapter subfolders under the Solution subfolders. These are intended for the instructor only.

2.3. Chapter Subfolders

For convenience the circuit files are contained in chapter subfolders. The names of the chapter subfolders correspond to the chapter number of Electronic Devices in which the circuits are referenced. For example, Chapter 3 circuits are contained in the CHAP03 subfolder, Chapter 4 circuits in CHAP04, and so forth.

2.4 Circuit Types

Electronic Devices references four types of circuit files, corresponding to examples, figures, troubleshooting exercises, and troubleshooting problems. Circuits are organized in subfolders corresponding to the circuit category within the chapter subfolder, as shown in Figure 3:

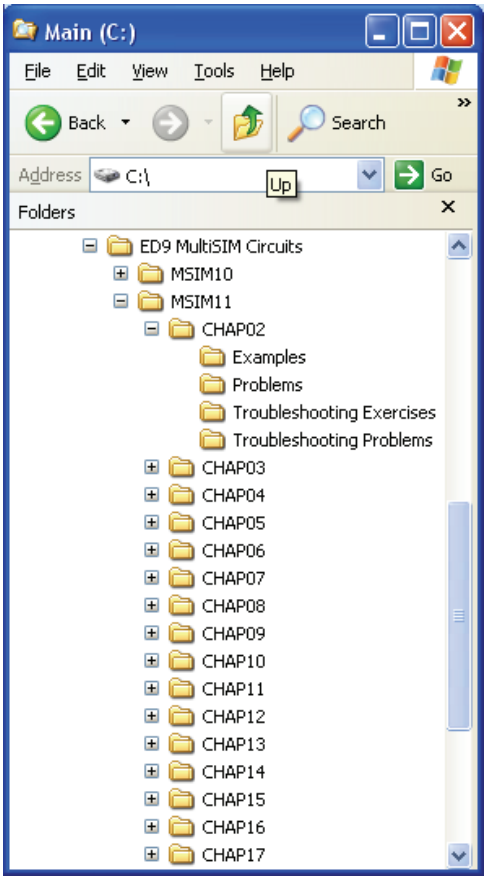


Figure 3: Category Folder Organization

3. Circuit Files

The MultiSIM circuit files consist of two basic types. The first type consists of simulation circuits. The second type consists of troubleshooting circuits.

3.1. Simulation Circuits

Simulation circuits are circuits intended to demonstrate or illustrate some principle of digital circuit theory. These circuits require the student to apply an input stimulus and observe the behavior of the circuit using an oscilloscope, multimeter, Bode plotter, or other monitoring device.

3.2. Troubleshooting Circuits

Troubleshooting circuits require the student to select some means of applying inputs and observing outputs. In addition, the student is expected to use the input and output signals to isolate and identify some hidden fault in the circuit or verify that the circuit is operating correctly.

3.3. Circuit Restrictions

All hidden faults in troubleshooting circuits are password-protected. Passwords for circuit restrictions are provided in the Electronic Devices Instructor's Resource Manual (IRM).

4. Circuit File Simulating Tips and Techniques

This section contains some guidelines for using MultiSIM to simulate the circuits provided for Electronic Devices. It is not and is not intended to be a general tutorial for using MultiSIM. For more information on using MultiSIM the student and instructor are referred to the MultiSIM User's Manual.

4.1. MultiSIM 10 and 11 vs. Earlier Versions of MultiSIM

The user with access to different versions of Multisim may notice that circuits that circuits may simulate differently or fail to simulate correctly under differenct versions of Multisim. There are some things that the user may try to correct the problem.

1) Install the latest Multisim upgrade or patch. These may be found at http://www.ni.com on the World Wide Web.

2) Use the Convergence Assistant to automatically adjust the simulation settings.

3) If possible, rebuild the circuit from scratch using Multisim 10 or Multisim 11 components, or contact Technical Support at Electronics Workbench. All the circuits contained on the CD have been tested to ensure that they simulate properly. All circuits were tested using Windows XP SP3 on an AMD Phenom 9500 2.2 GHz Quad Core Processor system with 4 GB of memory. Systems with other processors, system memory, or operating systems may result in slower or faster simulation times for some circuits.

4.2. Simulation Options

As indicated previously MultiSIM offers a number of options for supplying inputs and monitoring outputs in circuits. While there is no "right" or "wrong" way to simulate a circuit, the purpose of the simulation and the nature of the circuit often favor a particular approach. For circuits in which you desire to know some static value of an AC value (such as the magnitude of a voltage) a multimeter and RMS measurements are generally preferred. For dynamic (transient or time-varying) values and oscilloscope and peak or peak-to-peak measurements are usually better. Similarly, when you wish to observe the response to a single input change, such as a step input voltage, a switch is usually a better choice than a square wave generator.